# The Market's Hidden Signals: Using Biometric Data to Predict Financial Bubbles

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## Introduction

### Emotions influence decision-making.

- Electrodermal activity and heart rate changes are involved in financial choices [1].

### Risk preferences shift based on the context and individual experiences.

- Repeated exposure to risky tasks change individuals' displayed risk preferences [2, 4].
- Risk aversion varies by individual differences in personality beyond age or sex effects [3, 5].

### Hypotheses

- Skin conductance correlates with earnings.
- Emotional responses before decisions predict stock performance.
- Increased risk aversion correlates with larger market bubbles.

## Methodology

### Overview

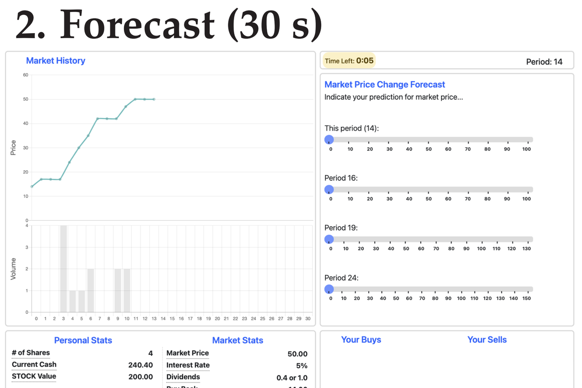
- $N = 358$ participants across 19 markets.
- 2–5 in-person and 15-18 online per session.
- Physiological data collected via Empatica E4 for in-person participants.

### Stock Trading Task

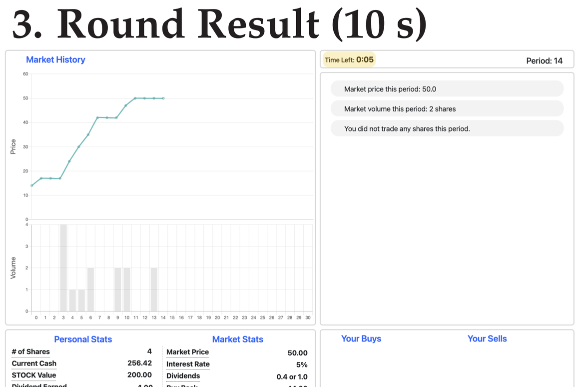
- Start with 4 stocks and 140 units of cash.
- Cash earns 5% interest per period across 30 periods.
- Option to trade a risky asset with dividends of 0.4 or 1, each with equal probability [6, 7].

**1. Order Submission (20 s)**

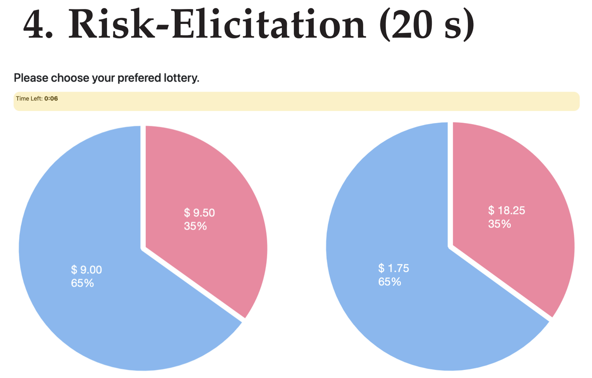
**2. Forecast (30 s)**



**3. Round Result (10 s)**



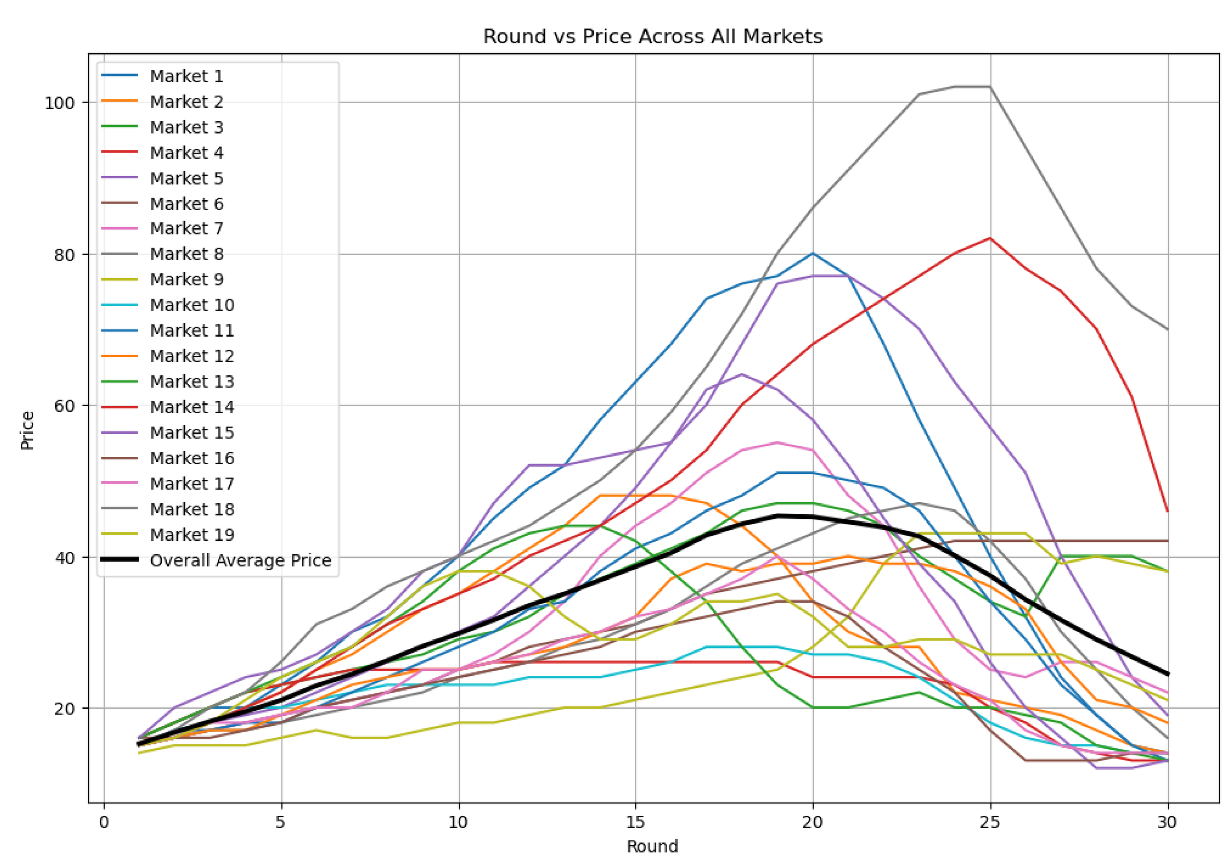
**4. Risk-Elicitation (20 s)**

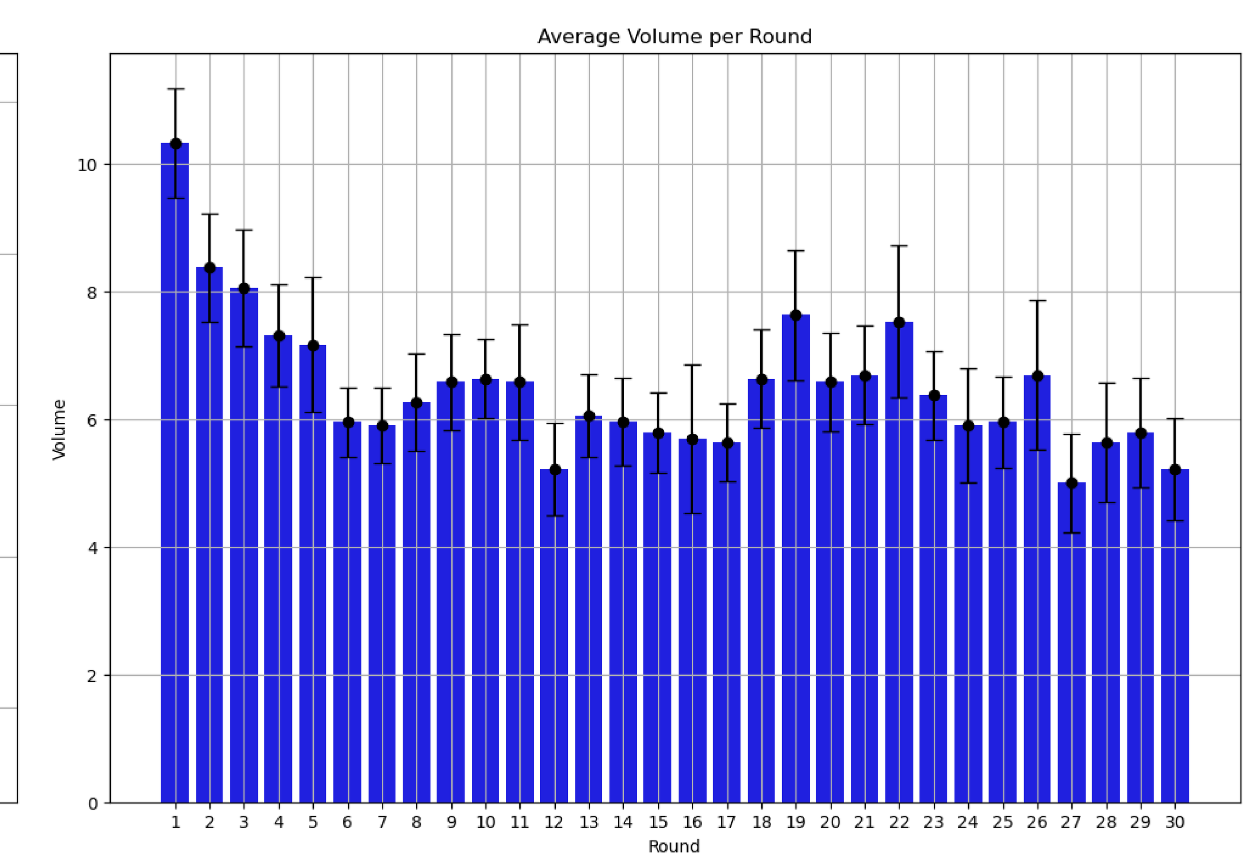


## Preliminary Results

### Market Summary

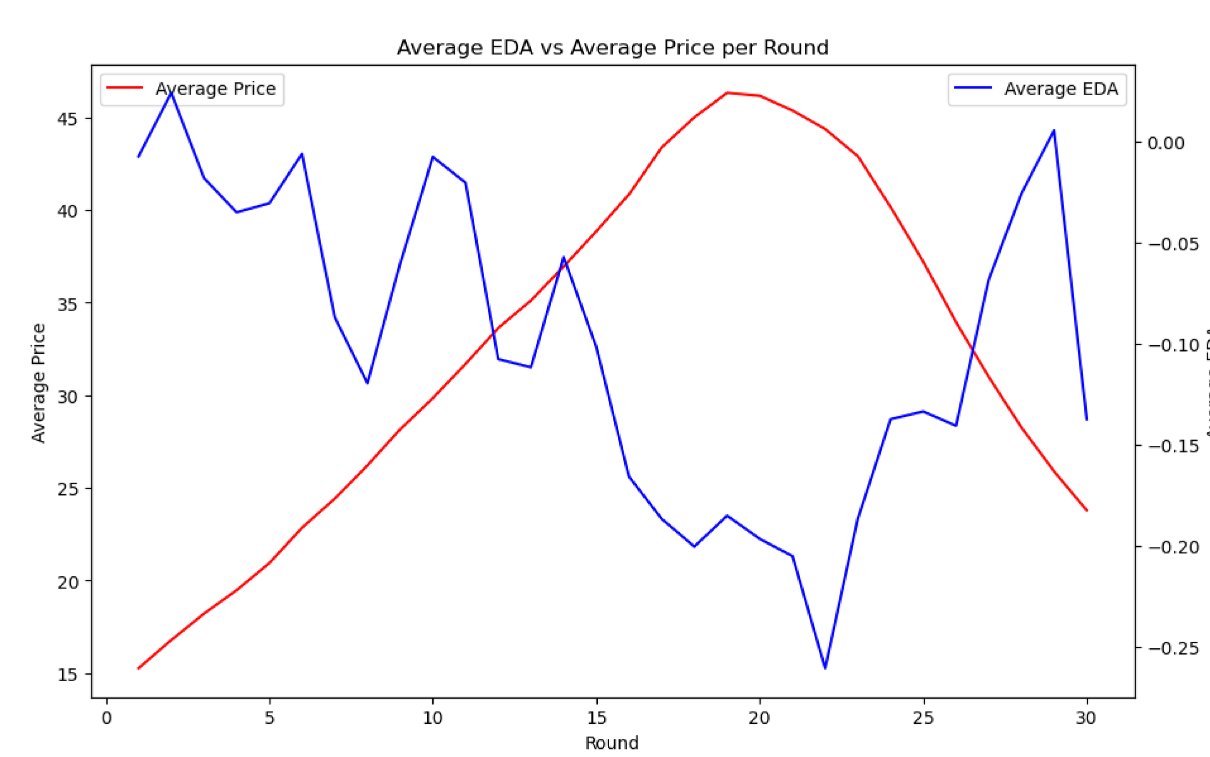
- Most markets display an initial increase in prices, followed by a peak in the latter half, and then a decline – aligning with the typical lifecycle of an asset bubble.
- Initially, trading volume is high. As the rounds progress, there is a decline in volume, reaching a low point around round 12.

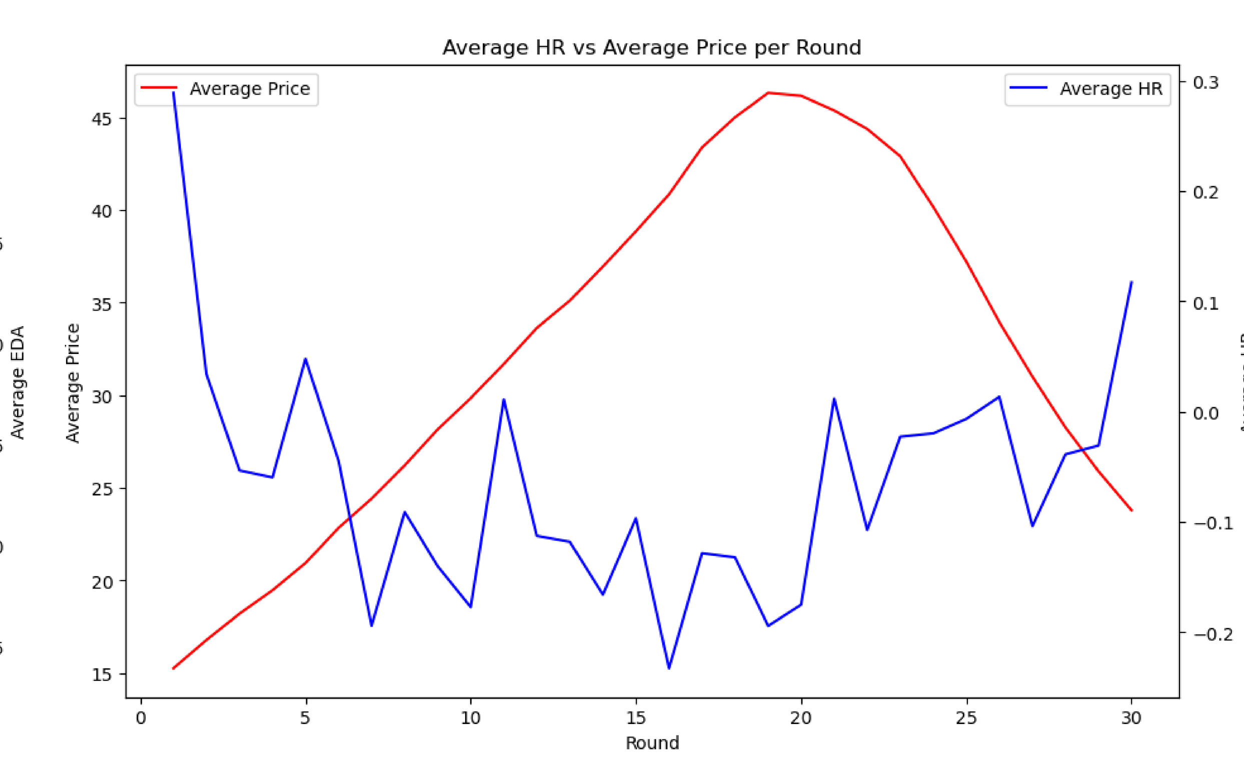


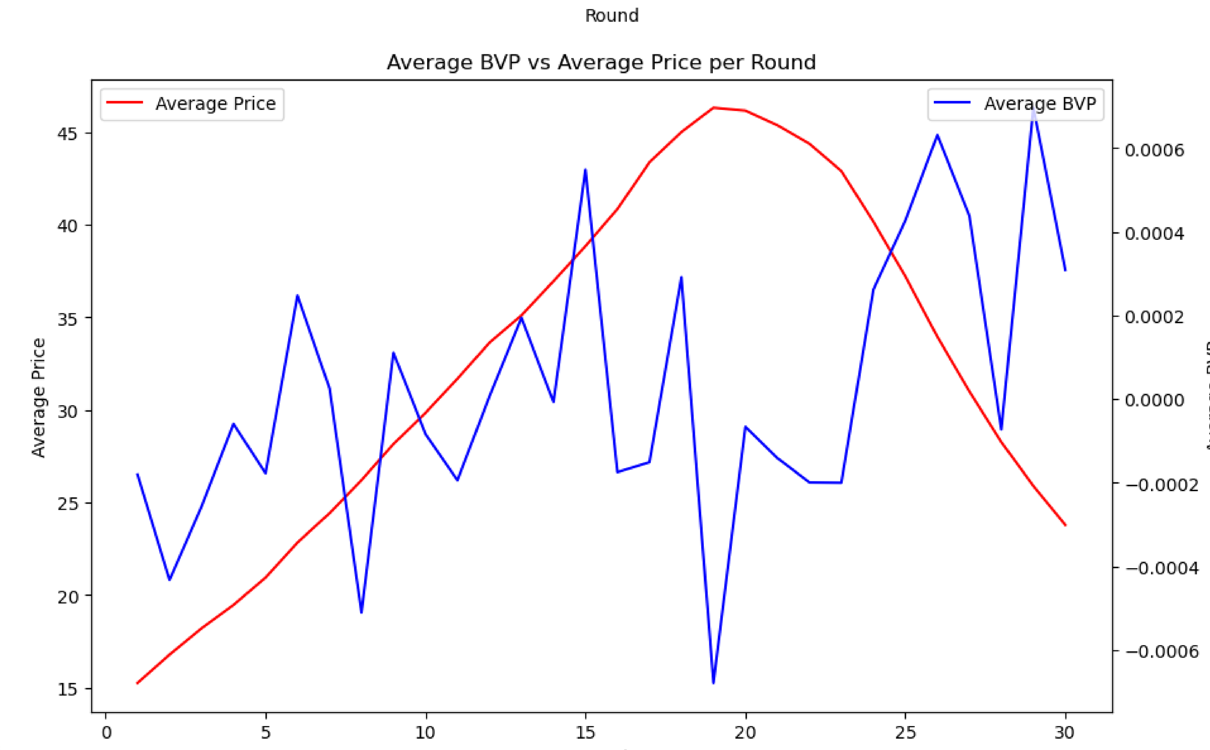



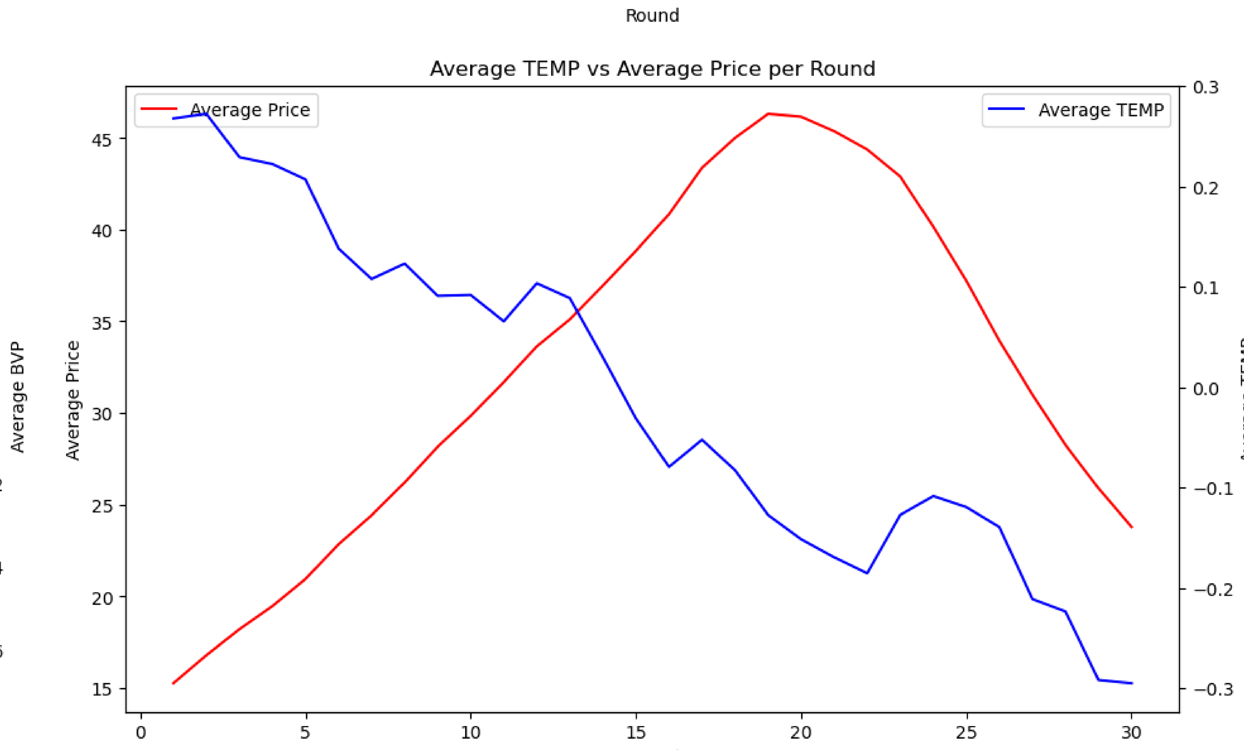

### Biometric Responses

- Electrodermal activity (EDA), heart rate (HR), blood volume pulse (BVP), and skin temperature (TEMP) overlay average price per round.




## Discussion

### Biometric Data Analysis

- Inverse relationship between EDA and HR versus price suggest emotional arousal aligning with market behavior.
- Lowest average value of BVP at price peak suggests reduced physiological stress, potentially due to market overconfidence.
- General decrease of TEMP may reflect diminishing emotional engagement or adaptation to market fluctuations.

### In Progress

- Moving averages will be used to smooth out fluctuations in biometric data and reveal trends more clearly.
- Use a fixed effect model to analyze how independent variables (market, PCA, biometric data) affect the dependent variable (average returns).
- Analyze forecasts with various ML models.

### Future Implications

- Potential for using physiological measures to predict market turning points and improve bubble management strategies.

The preliminary results suggest that biometric data may be correlated with stock returns and possibly predictive of bubble crashes.

## Acknowledgements

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## References

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